

CSE110 Spring 2024 Lab Exam - 03 Tentative Solutions and Rubrics

In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A]

SOLUTION

```
import java.util.Scanner;

public class SecondMaxPairSum {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        // Initialize max and second max with first two pairs
        int maxSum = arr[0] + arr[1];
        int secondMaxSum = arr[0] + arr[2];

        // Swap if needed
        if (secondMaxSum > maxSum) {
            int temp = maxSum;
            maxSum = secondMaxSum;
            secondMaxSum = temp;
        }

        // Loop through all valid pairs
        for (int i = 0; i < n - 1; i++) {
            for (int j = i + 1; j < n; j++) {
                int sum = arr[i] + arr[j];

                if (sum > maxSum) {
                    secondMaxSum = maxSum;
                    maxSum = sum;
                } else if (sum > secondMaxSum && sum < maxSum) {
                    secondMaxSum = sum;
                }
            }
        }

        // Output
        System.out.println("Second Maximum pair sum: " + secondMaxSum);
    }
}
```

[SET-B]

SOLUTION

```
import java.util.Scanner;
public class SecondMinPairSum {

    // Method to find the second minimum pair sum
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        // Initialize with the first two pairs
        int minSum = arr[0] + arr[1];
        int secondMinSum = arr[0] + arr[2];

        // Ensure minSum < secondMinSum
        if (secondMinSum < minSum) {
            int temp = minSum;
            minSum = secondMinSum;
            secondMinSum = temp;
        }

        for (int i = 0; i < n - 1; i++) {
            for (int j = i + 1; j < n; j++) {
                int sum = arr[i] + arr[j];

                if (sum < minSum) {
                    secondMinSum = minSum;
                    minSum = sum;
                } else if (sum > minSum && sum < secondMinSum) {
                    secondMinSum = sum;
                }
            }
        }

        System.out.println("Second Minimum pair sum: " +
secondMinSum);
    }
}
```

[CSE110 Spring 2024 Lab Exam - 0X Tentative Solutions and Rubrics](#)

RUBRIC

SL	Points to meet	Marks (10)
	Proper Java Syntax	1
	Correctly take user input	1.5
	Initialization of max/min and second max/min	1.5
	Correct nested loop logic	2
	Correct sum calculation	1
	Updating second max/min correctly	3
Total		10